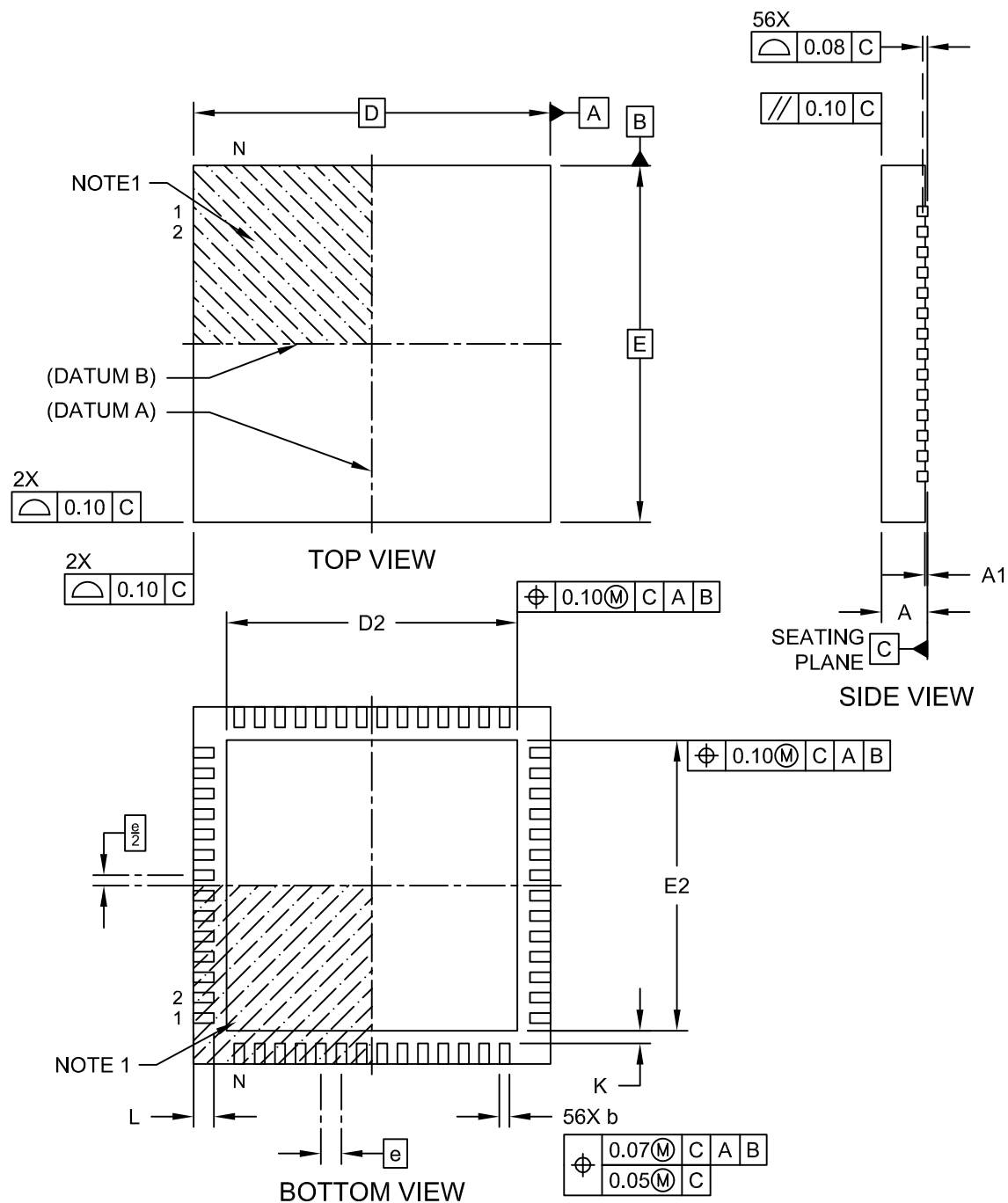


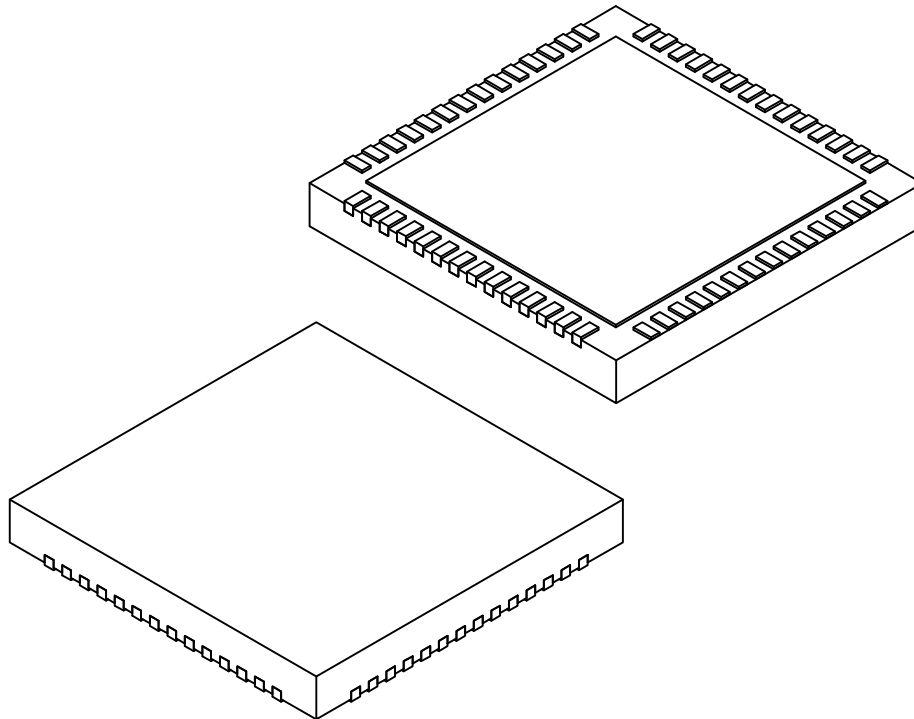
**56-Lead Very-Thin Quad Flatpack, No Lead Package (SPB) - 7x7x0.9 mm Body [VQFN]  
With 5.7x5.7mm Exposed Pad; Atmel Legacy GPC ZGM**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



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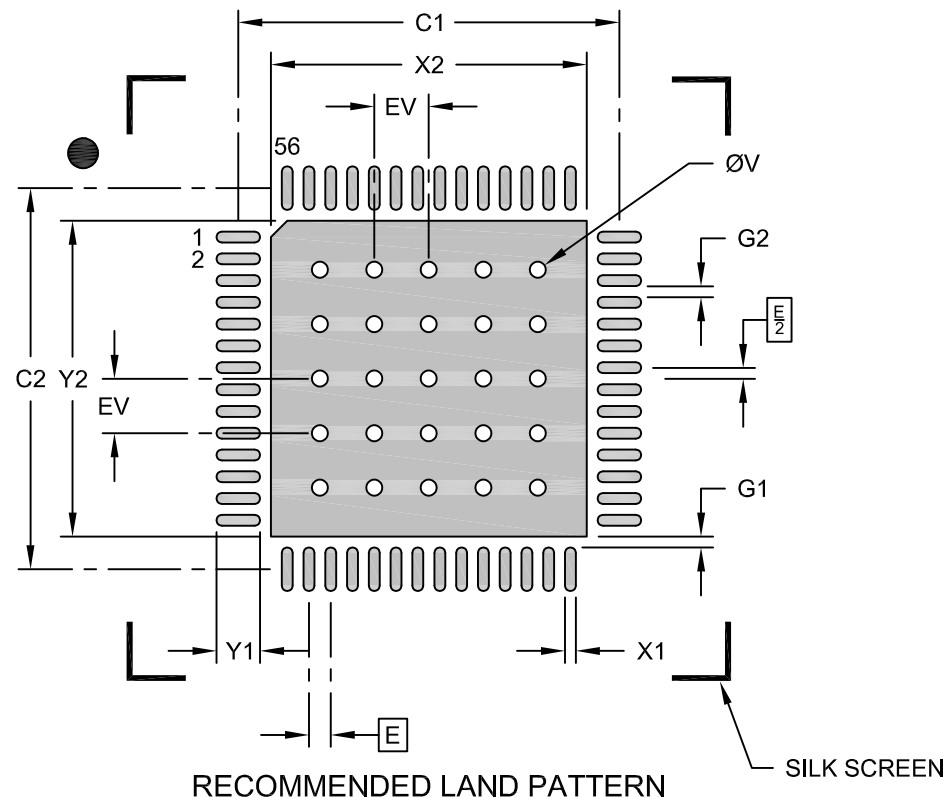
| Units                   |    | MILLIMETERS |      |      |
|-------------------------|----|-------------|------|------|
| Dimension Limits        |    | MIN         | NOM  | MAX  |
| Number of Terminals     | N  | 56          |      |      |
| Pitch                   | e  | 0.40 BSC    |      |      |
| Overall Height          | A  | 0.80        | 0.85 | 0.90 |
| Standoff                | A1 | 0.00        | -    | 0.05 |
| Overall Length          | D  | 7.00 BSC    |      |      |
| Exposed Pad Length      | D2 | 5.60        | 5.70 | 5.80 |
| Overall Width           | E  | 7.00 BSC    |      |      |
| Exposed Pad Width       | E2 | 5.60        | 5.70 | 5.80 |
| Terminal Width          | b  | 0.15        | 0.20 | 0.25 |
| Terminal Length         | L  | 0.35        | 0.40 | 0.45 |
| Terminal-to-Exposed-Pad | K  | 0.25        | -    | -    |

**Notes:**

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
  - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
  - REF: Reference Dimension, usually without tolerance, for information purposes only.

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| Units                            |    | MILLIMETERS |      |      |
|----------------------------------|----|-------------|------|------|
| Dimension Limits                 |    | MIN         | NOM  | MAX  |
| Contact Pitch                    | E  | 0.40 BSC    |      |      |
| Center Pad Width                 | X2 |             |      | 5.80 |
| Center Pad Length                | Y2 |             |      | 5.80 |
| Contact Pad Spacing              | C1 |             | 7.00 |      |
| Contact Pad Spacing              | C2 |             | 7.00 |      |
| Contact Pad Width (Xnn)          | X1 |             |      | 0.20 |
| Contact Pad Length (Xnn)         | Y1 |             |      | 0.80 |
| Contact Pad to Center Pad (Xnn)  | G1 | 0.20        |      |      |
| Contact Pad to Contact Pad (Xnn) | G2 | 0.20        |      |      |
| Thermal Via Diameter             | V  |             | 0.30 |      |
| Thermal Via Pitch                | EV |             | 1.00 |      |

**Notes:**

- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23416 Rev A